

Valid Inference with Synthetic Data via Task Exchangeability

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June 13, 2026

Abstract

We present **exchangelib**, a Python package that implements the full statistical pipeline for *task-exchangeability* inference with synthetic data. The library follows a clean **src/** layout, provides a minimal **pyproject.toml**, and exposes a public API of the form **from exchangelib.<module> import <Class>**. Core components include a formal definition of exchangeability between real and synthetic tasks, kernel-based density estimation, bias-correction procedures, and a suite of diagnostics that verify the theoretical guarantees of the pipeline. The implementation is deliberately reference-level: no empirical benchmarks are reported, and all quantitative claims are deferred to future work. The accompanying test suite checks structural properties (e.g., proper handling of package imports, consistency of kernel estimators, and invariance of bias-corrected statistics under simulated exchangeable transformations) rather than performance metrics. This paper describes the design, algorithms, and validation strategy of the library, positioning it as a reproducible baseline for researchers exploring valid inference with synthetic data.

1 Introduction

Synthetic data are increasingly used to protect privacy, accelerate experimentation, and enable reproducible research. However, statistical inference based on synthetic data is only valid when the synthetic generation mechanism respects certain probabilistic relationships with the original task. *Task exchangeability* formalises this relationship: a synthetic task is exchangeable with the real task if the joint distribution of observed data and synthetic data is invariant under permutations of the task index. Under this assumption, unbiased estimators and valid confidence intervals can be derived using kernel density methods and bias-correction techniques.

Despite its theoretical appeal, task-exchangeability has not been operationalised in an open-source software package. Existing tools for synthetic data analysis either ignore exchangeability or provide ad-hoc corrections that lack a unified statistical foundation. We therefore develop **exchangelib**, a reference implementation that encodes the full pipeline—from definition of exchangeability, through kernel estimation, to inference and diagnostics—within a modular, test-driven Python package.

2 Related Work

The literature on synthetic data inference includes early work on posterior predictive checks [?], recent developments in differential privacy [?], and kernel-based approaches for density estimation [?]. Task exchangeability was introduced in the context of causal inference [?] and later specialised to synthetic data [?]. Existing software such as **synthetic-data** [?] and **privacy-tools** [?] provide utilities for data generation but do not implement the formal exchangeability framework. Our work builds directly on the theoretical foundations of ? and integrates the kernel methods of ? with the bias-correction strategies of ?.

3 Methodology

3.1 Statistical Framework

Let $\mathcal{T} = \{T_1, \dots, T_K\}$ denote a collection of tasks, where T_1 is the real task and $T_{2:K}$ are synthetic tasks generated by a known mechanism $\mathcal{G}$. The tasks are *exchangeable* if for any permutation π of $\{1, \dots, K\}$,

$$P((X_{T_1}, Y_{T_1}), \dots, (X_{T_K}, Y_{T_K})) = P((X_{T_{\pi(1)}}, Y_{T_{\pi(1)}}), \dots, (X_{T_{\pi(K)}}, Y_{T_{\pi(K)}})).$$

Under exchangeability, the marginal distribution of the real task can be estimated by aggregating over all tasks, yielding an unbiased estimator of any functional $\theta = \mathbb{E}[f(X, Y)]$.

3.2 Kernel Estimation

We employ a multivariate kernel density estimator (KDE) $\hat{p}_h$ with bandwidth matrix h :

$$\hat{p}_h(z) = \frac{1}{K} \sum_{k=1}^K K_h(z - Z_{T_k}),$$

where $Z_{T_k} = (X_{T_k}, Y_{T_k})$ and K_h is a product of Gaussian kernels. The bandwidth is selected via a plug-in rule that balances bias and variance under the exchangeability assumption.

3.3 Bias Correction

Finite-sample bias arises from the kernel smoothing and from the synthetic generation process. Following ?, we construct a bias-corrected estimator

$$\tilde{\theta} = \hat{\theta} - \hat{b},$$

where $\hat{b}$ is obtained by a leave-one-out jackknife across tasks. The correction is provably unbiased when the tasks are exchangeable.

3.4 Diagnostics

Diagnostics verify the exchangeability assumption by testing permutation invariance of empirical moments and by assessing the stability of the KDE under task resampling. These checks are implemented as statistical tests (e.g., permutation tests) and visual tools (e.g., QQ plots).

4 Implementation

The package follows a conventional `src/` layout:

```
exchangelib/
src/
  exchangelib/
    __init__.py
    bias_correction.py
    cli.py
    diagnostics.py
    exchangeability.py
    inference.py
    kernel.py
    repository.py
```

```

    utils.py
tests/
    ... (pytest suite)
pyproject.toml
.gitignore

```

Modules

- `exchangeability.py` implements the formal definition, provides a `TaskExchangeability` class, and supplies permutation utilities.
- `kernel.py` contains the `KernelEstimator` class with methods for KDE evaluation, bandwidth selection, and gradient computation.
- `bias_correction.py` defines `BiasCorrector`, which performs jackknife bias estimation and returns corrected estimates.
- `diagnostics.py` provides diagnostic tests (e.g., `permutation_test`, `stability_plot`) and utilities for visual inspection.
- `repository.py` abstracts data loading, supporting both real and synthetic task datasets.
- `cli.py` implements a minimal command-line interface for end-users to run the pipeline on a directory of tasks.
- `utils.py` contains helper functions (e.g., distance metrics, logging).

All public symbols are re-exported in `exchangelib/`*init.py* to enable imports such as `from exchangelib.kernel import KernelEstimator`.

Testing Strategy The test suite (built with `pytest`) verifies:

1. Correct package importability and namespace exposure.
2. Mathematical consistency of the KDE (e.g., integration to one on synthetic grids).
3. Invariance of `TaskExchangeability` under permutations of task indices.
4. Proper functioning of the jackknife bias estimator (the corrected estimate equals the uncorrected estimate when synthetic tasks are identical).
5. Diagnostic functions raise appropriate warnings when exchangeability violations are simulated.

No performance benchmarks or empirical coverage guarantees are reported; the tests focus on structural correctness and adherence to the theoretical pipeline.

5 Validation & Limitations

The current validation is limited to the synthetic test harness described above. The test suite demonstrates that the implementation respects the mathematical definitions of exchangeability, that kernel estimators behave as expected on controlled data, and that bias correction reduces systematic deviations in the simulated setting. However, the library has not been applied to real-world synthetic datasets, and no quantitative assessment of statistical power, confidence-interval coverage, or computational efficiency has been performed. Consequently, the practical robustness of the methods under model misspecification, high dimensionality, or limited synthetic sample sizes remains an open question.

6 Conclusion and Future Work

We have introduced `exchangelib`, a reference Python package that operationalises task-exchangeability inference with synthetic data. The library provides a modular, well-tested implementation of the core statistical components|exchangeability definition, kernel density estimation, bias correction, and diagnostics|while exposing a clean public API. At present, validation is confined to unit tests that confirm theoretical properties; empirical evaluation on realistic synthetic data, scalability studies, and integration with privacy-preserving generation mechanisms are deferred to future work. Extending the package with adaptive bandwidth selection, support for non-Gaussian kernels, and comprehensive benchmarking will further solidify its role as a baseline for the community.